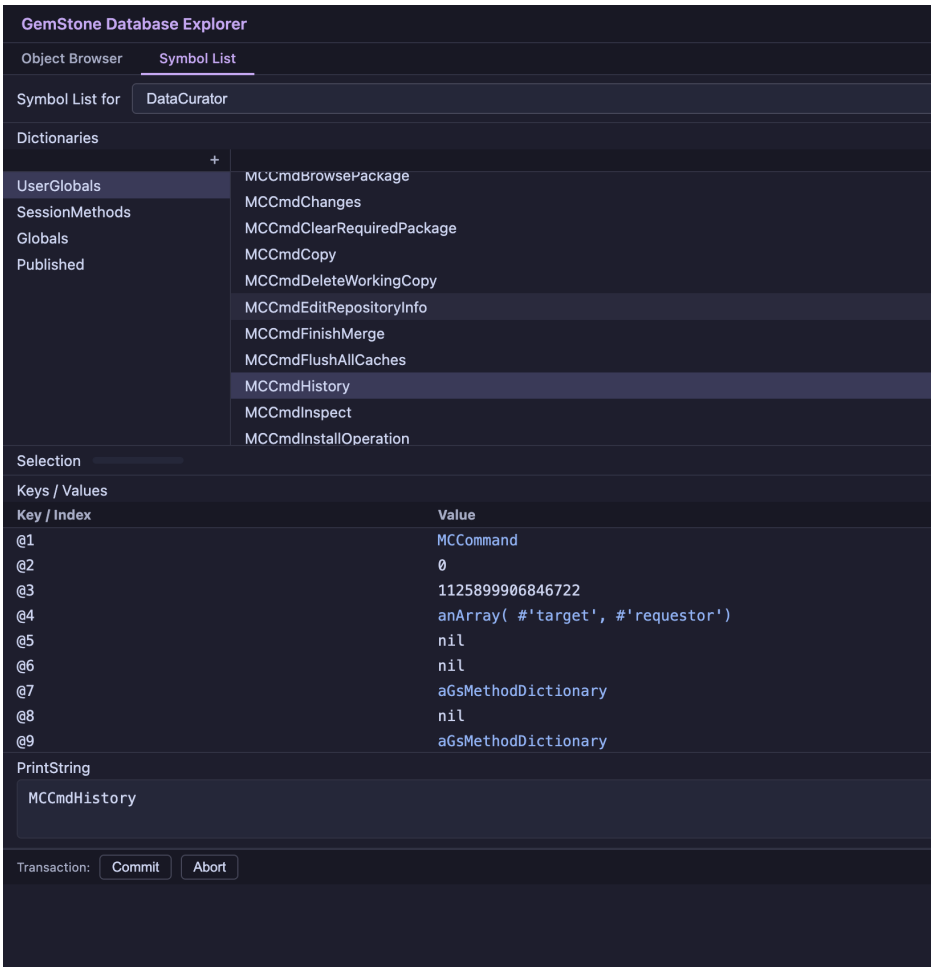
GemStone Database Explorer

A Python/Flask web application for browsing and inspecting objects in a GemStone/S object database. Port of maglev-database-explorer-gem using gemstone-py as the backend.



GemStone Database Explorer

Features

- **Object Browser** — navigate the object graph from well-known roots (UserGlobals, Globals, System); inspect instance variables, dictionary entries, and collection elements; evaluate Smalltalk expressions in the context of any object
- **Symbol List Browser** — browse GemStone symbol lists by user and dictionary; view key/value pairs and printStrings; add and remove dictionaries and entries
- **Transaction control** — Commit and Abort buttons on both tabs

- Batched GCI evaluation for fast object inspection (single `eval()` per view)

Requirements

- GemStone/S 64 3.x with GCI libraries accessible on `LD_LIBRARY_PATH` / `DYLD_LIBRARY_PATH`
- Python 3.11+
- gemstone-py

Installation

```
git clone https://github.com/unicompute/python-gemstone-database-
        explorer
cd python-gemstone-database-explorer
python3 -m venv .venv
.venv/bin/pip install -e .
```

Configuration

Set the following environment variables before starting:

Variable	Description	Example
`GEMSTONE`	Path to GemStone installation	`/opt/gemstone/GemStone64Bit3.7.5-arm64.Darwin`
`GS_USERNAME`	GemStone login username	`DataCurator`
`GS_PASSWORD`	GemStone login password	`swordfish`
`GEMSTONE_NRS`	Network Resource String for the Stone	`!tcp@localhost#server!gemstone`

See docs/configuration.md for full details.

Usage

```
.venv/bin/python-gemstone-database-explorer
# → GemStone Database Explorer running at http://127.0.0.1:9292/
```

Options:

```
--host HOST    Bind host (default: 127.0.0.1)
--port PORT    Port (default: 9292)
--debug        Enable Flask debug mode
```

Open `http://127.0.0.1:9292/` in a browser.

Documentation

- Configuration
- Architecture
- API Reference

License

MIT